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Roll No.....

EX-8301-GS

B.E. VIII Semester

Examination, December 2020

Grading System (GS)

Advanced Power Electronics

(Elective - III)

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1. Discuss the working of an isolated flyback regulator and derive the expression for RMS value of primary current.
2.
 - a) Explain the working principle of Buck Boost Converter with appropriate waveforms of each converter.
 - b) A boost regulator has an input voltage of 12V. The average output voltage is 15V with an average load current of 0.5A. The chopping frequency is 20 kHz. If $L = 200\text{mH}$ and $C = 400\text{ pF}$. Calculate the duty ratio and ripple current of the induction.
3.
 - a) Explain working of full bridge converter type switch mode dc power supply along with its applications and relative merits.
 - b) Discuss the concept of average model and state space model of converter.
4.
 - a) For the SVM technique in inverters explain the terms space transformation and space vectors.

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- b) Explain SPWM used in PWM inverters. Explain undermodulation and over modulation modes of working.
5. a) With the help of circuit diagram explain the working of bidirectional voltage controller. Draw the waveforms of output voltage for conduction angle $\alpha = 60^\circ$ and $\alpha = 90^\circ$.
- b) A three phase full wave bidirectional voltage controller supplies λ connected resistive load of 10Ω and input supply is 200 V, 50 Hz. For the delay angle $\alpha = 60^\circ$, sketch the output voltage waveform. Determine the Fourier expression of output voltage and comment on harmonic contents.
6. a) What is ac chopper? Describe the modeling of ac chopper.
- b) Explain harmonics control technique using symmetrical waveform pattern for ac chopper.
7. Explain the principle of ZVS converter. Describe a ZVS resonant converter with appropriate circuit and waveform.
8. Write Short Notes (Any two) :
- a) CUK converters
 - b) ZCT DC converters
 - c) Generalized switching cell
 - d) Soft switching advantages

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